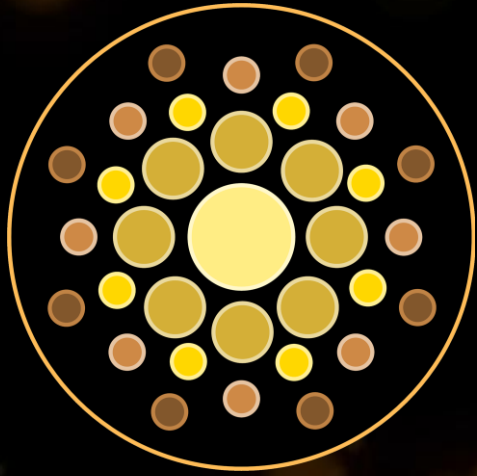




Research Net

Accelerate research and product development with an agentic automation platform for the complete product lifecycle

<https://www.researchnet.ai/>

India Hub Presentation

February 4th, 2026



The PROBLEM



Manual Analysis is
Slow and subjective



Professional Analytics
are Inaccessible

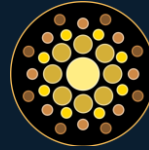


Feedback Lacks a
Visual Representation



Critical Match Insights
are Lost





Product Steps



Ultralytics and YOLO v8

Create the starting model

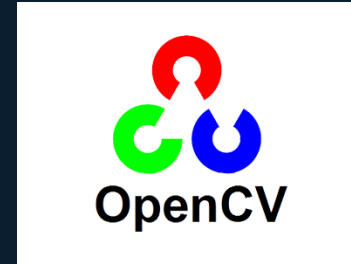
- most optimized tradeoff between accuracy and speed
- three main models: v8n, v8s, v8m
- works directly with COCO datasets right away
- can withstand live match analysis



Kaggle Datasets

Training the Model

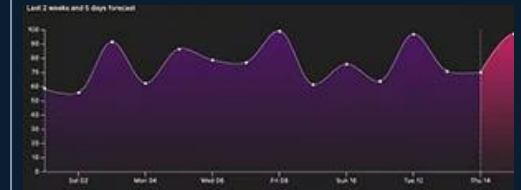
- the data set used right now has around 1800 images
- encompasses diverse match conditions
- captures authentic scenarios
- preparing models for motion blur, swing etc.



Live Video Integration

Using OpenCV to prepare model for live games

- OpenCV will help further automate the frame-by-frame output
- mapping video frames with field dimensions/player builds etc
- better trajectory/trail detection



Building the Product

Allowing players to track their stats post-match

- intuitive GUI/dashboard for players and coaches to navigate easily
- advance metrics: more than just line and length for example
- every statistic should have the video it is referencing accessible/skeletal tracking



Overarching Methodology

- Started by using v8n but later realized that is not sufficient for detecting small blurry objects like cricket balls
- Now am using a more sophisticated approach in the [multi-strategic.py](#) file as it not only uses the superior v8m model, it checks for color shape and size, allowing for more consistent detection
- The [train-model.py](#) file is how I will eventually start training the model connecting to Kaggle to create a model that is specifically trained for cricket balls, existing COCO classes only recognize “sports balls”
- The Java files are used to help out in places Python struggle such as cross-platform GUI Framework, and to build core functionality

```
> .vscode
> bin
> lib
> output
✓ src
  J YoloJavaHelpers.java
  J YoloProgram.java
  multi-strategic.py
  README.md
  test-01.py
  test-02.py
  train_model.py
  v8s-improved.py
  yolov8m.pt
  yolov8n.pt
  yolov8s.pt
```



```
YOLO_CONFIG = {
    "model": "yolov8m.pt",
    "class_id": 32,          # COCO sports ba
    "conf": 0.05,           # Low threshold
    "iou": 0.3,
    "imgsz": 1920,
    "tracker": "bytetrack.yaml",
}
```

```
all_dets = []
for cx, cy, conf, bbox in yolo_dets:
    all_dets.append({"center": (cx, cy), "conf": conf, "strategy": "yolo", "bbox": bbox})
for cx, cy, conf, bbox in color_dets:
    all_dets.append({"center": (cx, cy), "conf": conf, "strategy": "color", "bbox": bbox})
for cx, cy, conf, bbox in motion_dets:
    all_dets.append({"center": (cx, cy), "conf": conf, "strategy": "motion", "bbox": bbox})

if not all_dets:
    return []

used = [False] * len(all_dets)
clusters = []

for i in range(len(all_dets)):
    if used[i]:
        continue
    used[i] = True
    cluster = [all_dets[i]]

    for j in range(i + 1, len(all_dets)):
        if used[j]:
            continue
        dx = all_dets[i]["center"][0] - all_dets[j]["center"][0]
        dy = all_dets[i]["center"][1] - all_dets[j]["center"][1]
        if (dx*dx + dy*dy) < MERGE_RADIUS * MERGE_RADIUS:
            cluster.append(all_dets[j])
            used[j] = True

    clusters.append(cluster)

results = []
for cluster in clusters:
```



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Coaching App in Action

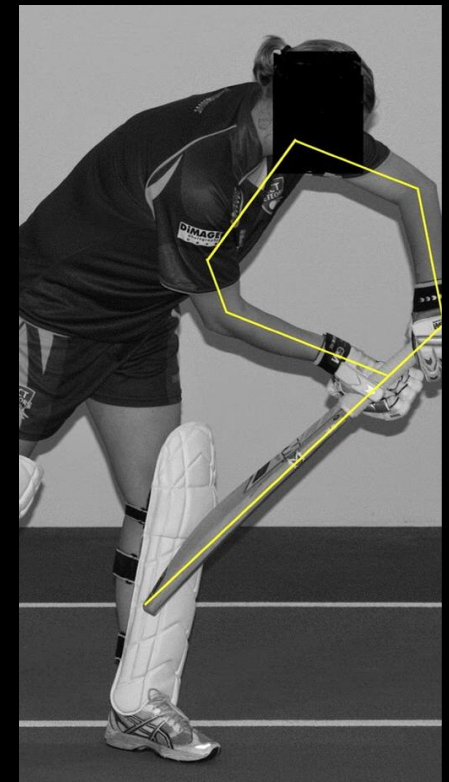
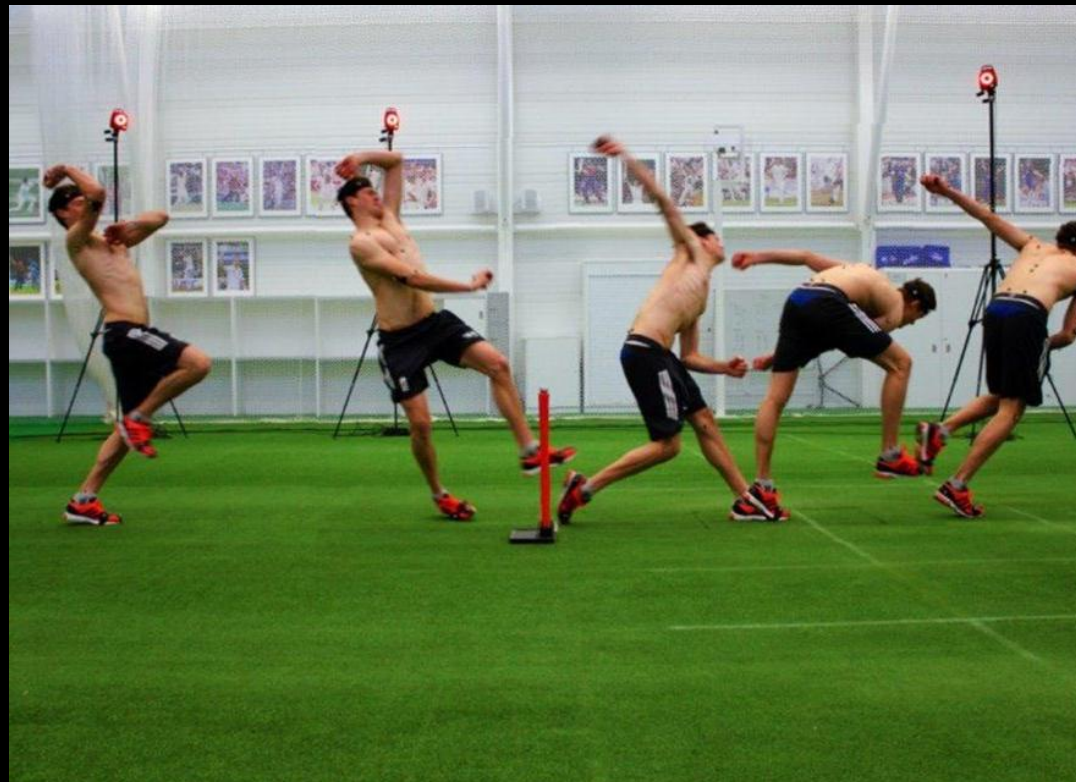
- YOLO v8 successfully identifies and tracks the cricket ball in real time
- Bounding box accurately follows ball movement frame by frame
- Detection is stable across positions and backgrounds
- When the model is further trained with the Kaggle Data Set it will be much stronger because it will be able to identify the seams of the cricket ball etc
- The next steps for this app is as follows:
 - Learning how to retrieve the ball trajectory
 - Calculating the point in which the ball bounces
 - Full GUI integration for a coach and player friendly interface





Next Steps: Studying Biomechanics

- Quality fast bowling has four distinct phases: the run-up, the delivery stride, releasing the ball, and the follow-through, all of which should be optimized fully
- In the past few decades, coaches have already discovered optimization techniques such as knee bracing and upper trunk flexion to generate the most torque
- Different types of deliveries and their usefulness can also be detected by analyzing the biomechanics of fast bowler's wrist movement, and their grip of the seam on the ball
- It is also important to note that the most textbook biomechanical way might not be the most optimal for everyone but small tweaks can be made based on player preferences or game situation

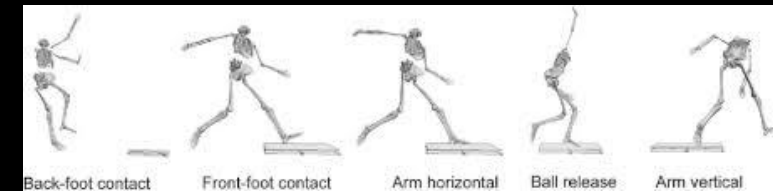
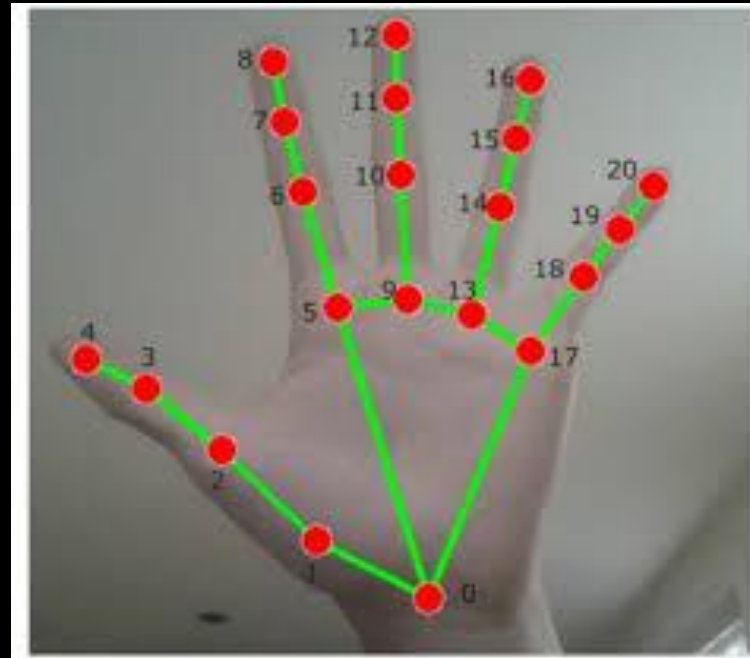
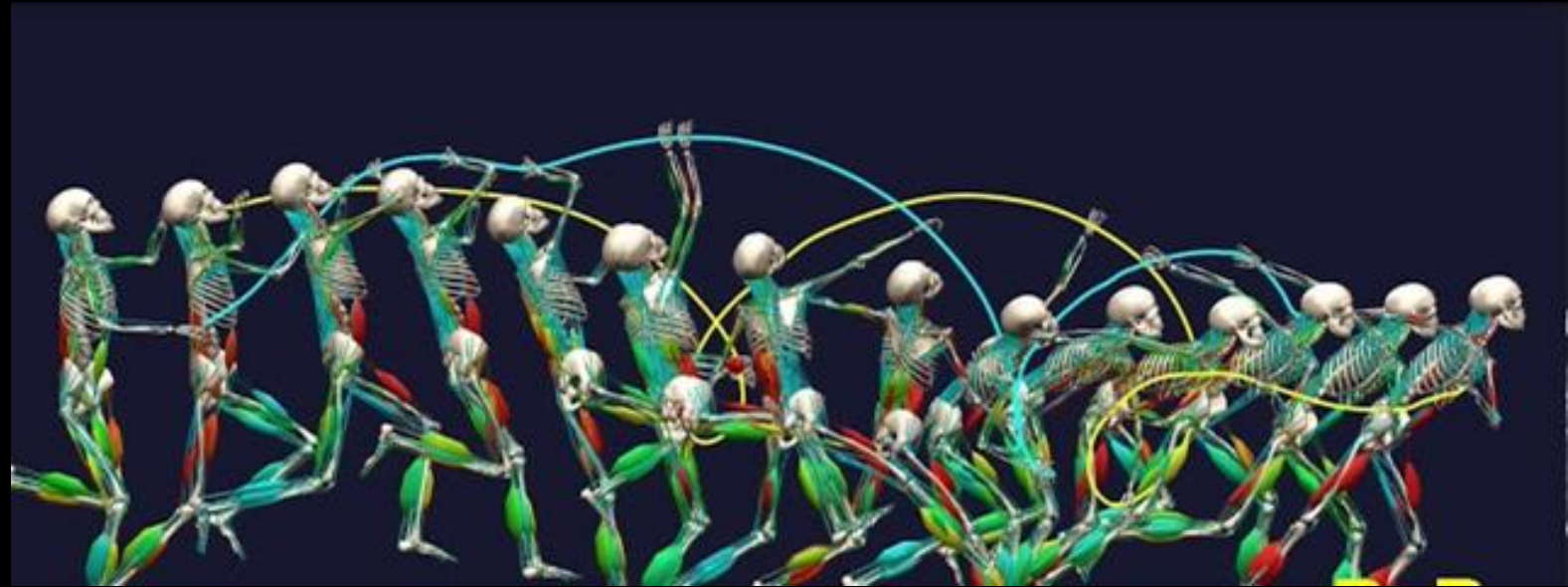




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Using Biomechanics to Analyze Skeletal Movements

- Libraries such as MediaPipe can extract around 20 joints (critical skeletal points as they call them) just within the hand. For one's entire body there can likely be close to 50 total points allowing the player and coach to analyze every specific muscle and joint movement
- The app will be designed to overlay these lines to gain real-time analytics and angles. This is also important because you want to ensure the app only recognizes movements that are possible for the human body.
- Dynamic Time Warping (DTW) algorithms shall be used in the product as well to align a player's video which they want to review into a professional database, allowing for accurate comparison, and insightful analytics.



The SOLUTION



Statistical Analysis
Rooted in Data



A Product Accessible
to even Club Players



A User-Friendly Easily-
Navigated Interface



Easy to Receive Post-
Match Insights



Works Cited:

Abraham, Alan, Simon A. Feros, and Aaron S. Fox. "Exploring the Accuracy of OpenCap for Three-Dimensional Analysis of Cricket Bowling." *International Journal of Sports Science & Coaching*, vol. 20, no. 6, 2025, pp. 2571-86, <https://doi.org/10.1177/17479541251348081>.

Felton, Paul. "Cricket Bowling Biomechanics | Dr Paul Felton." YouTube, uploaded by Stuart McErlain-Naylor.

Marwah, Nalin. "Cricket Fast Bowling Optimization Using Machine Learning Pose Estimation Modeling." *Research Archive of Rising Scholars*, preprint, 2025.

McErlain-Naylor, Stuart. "Cricket Batting Biomechanics | Dr Stuart McErlain-Naylor." YouTube, uploaded by Stuart McErlain-Naylor, [Date not provided in source transcript],.

Siddiqui, Hafeez Ur Rehman, et al. "Enhancing Cricket Performance Analysis with Human Pose Estimation and Machine Learning." *Sensors*, vol. 23, no. 15, art. 6839, 1 Aug. 2023, <https://doi.org/10.3390/s23156839>.

Thank You

Maake Asante Dhanyavadagalu Shukria
Vinaka Kaitos Maana Dankon Manana
감사합니다 Dank Je Dankscheen Kam Sah Hammida
Blagodaram Ngiyabonga Dziekuje Mauruuru Biyan
Juspaxar Arigato Chokrane Diolch i Chi Terima Kasih Matondo
நன்றி Bedankt Dakujem Grazas Mochchakkeram Tack
Ua Tsaug Rau Koj Dakujem Gracies Tingki
Děkuji Nirringrazzjak Hvala Gracías Gratias Tibi
Suksama Rahmat Matur Nuwun 谢谢 Di Ou Mèsi
Misaoatra Welalin Danke Kia Ora Kop Khun Khap Obrigado
Matur Nuwun 谢谢 xBala Merci Go Raibh Maith Agat
Salamat ขอบคุณคุณคุณ Najis Tuke Djierie Dieuf Eskerrik Asko